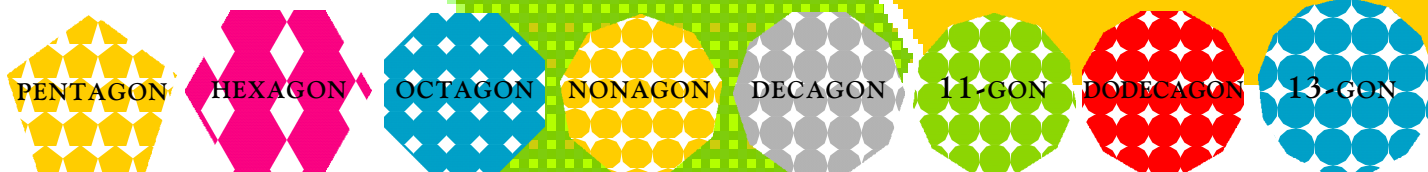


SHAPES *with many* SIDES

Polygons with many sides have Greek names based on the number of sides. Pentagons and hexagons, which have 5 and 6 sides, are the most common. Shapes with many more sides are rare and have strange names like “11-gon” and “13-gon”.



As the number of sides increases, polygons look more and more like circles. One way of describing a circle, therefore, is as a polygon with an *infinite* number of sides.



Nature's pentagons

Pentagons are rare in nature, but they do crop up in a few places.

Cut an apple in half and look at the seeds – they're arranged in a ring of five. Starfish and sea urchins have bodies with five parts arranged in a ring.

Do pentagons fit together?

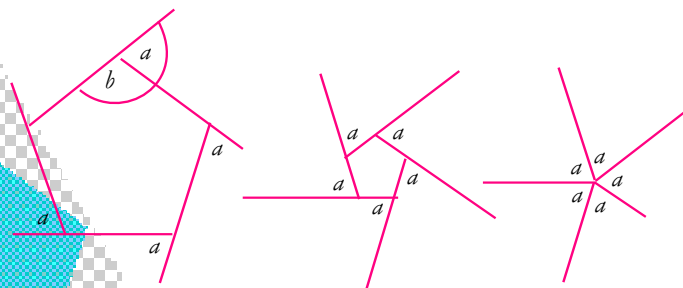
Pentagons won't tile a flat surface without gaps because their inner angles don't add up to 360° .

However, you can use a mixture of pentagons and hexagons to tile a curved, 3D surface. Have a look at a football, and you'll see that's exactly how they're made.



Adding the angles

Here's a clever way to work out the inside angles of a polygon. The diagrams show a pentagon, but it works for any polygon. First think about the outside angles (labelled a). If you imagine the shape shrinking down to a dot, it's clear that the outside angles add up to a full circle, which is 360° . So each outside angle must be a fifth of this, which is 72° . Now for the inside angles (b). The first diagram shows that a and b make a half circle, which is 180° . Since $a = 72^\circ$, b must be 108° .



TO MAKE A PENTAGON, cut a strip of paper about 3 cm (1 in) wide and tie it in a knot. As you slowly flatten and tighten the knot, it will form a pentagon. When you hold it to the light, you'll also see a 5-sided star (a pentagram)!

